

Look, what you are in for...

You would surely have read of smartappliances, smart thermostats, smart lighting systems and so on. But do you know that smart, autonomous gadgets are available for many more purposes such as the following?

- To water plants, depending on soil moisture, weather reports, etc
- To take orders and mix cocktails
- To monitor the condition of water in pools and autonomously initiate cleaning and refilling operations
- To automatically start cooking food as soon as you leave office

help them live independently. In the case of Eric Pebbles, an active college professor, who also suffers from spastic cerebral palsy due to oxygen deprivation at birth, a smarthome is a boon.

He roped in smart technology provider Smart Solutions to build an independent living space for him. After assessing his requirements, they came up with a detailed plan for the house, which included automated doors, lights, thermostats, shades and security cameras, which could be controlled through an app on his smartphone. This enabled Pebbles to control these equipment from anywhere in the house.

Smartlogic built into the system executes commands based on the situation without being explicitly given each instruction. When Eric is ready to go out, a one-touch command executes multiple tasks like unlocking and opening the door, switching off the air-conditioner and lights, opening the garage door and closing all these after he leaves.

In a Web report, Pebbles said, "You do not know how good it feels to be able to open my own front door and enjoy the view. Thanks to technology, I can." The best part is that all this magic has been worked into a normal apartment and not a specially-constructed one, so it is something anybody can try and implement for themselves.

Avoiding bugs

It is evident that as smarthome technologies gain momentum, these will serve a mixed purpose of convenience, comfort and necessity. In the

case of such critical applications, say, where a person's parameters or whereabouts are monitored for healthcare reasons, it is also important that the system should work without faults.

Understandably, researchers around the world are attempting to identify and overcome bugs in smart technologies. In one such research published by Juan Ye and team at School of Computer Science, University of St Andrews, Scotland, they observe that abnormal sensor events hamper the correct identification of critical (and potentially life-threatening) situations, and that existing learning, estimation and time based approaches are inaccurate and inflexible when applied to multiple people sharing a living space.

They also point out that sensor anomaly, which might be caused by people accidentally moving or dislodging sensors, can be even more problematic than broken sensors. This is because dislodged sensors could report values that are within a reasonable range but not accurate, so it would be difficult to find out that these are malfunctioning. Addressing this is, according to the team, more difficult than straightforward damage of sensors.

To solve this problem, the researchers have developed CLEAN, a knowledge-driven technique to detect anomalies in event-driven binary sensors. According to the authors, "CLEAN combines knowledge and statistical models by using well-defined knowledge as part of a clustering based outlier-detection technique."

The system does not rely on any training data or annotated data and is therefore not affected by any change in the occupants' routine activities. A flexible and dynamic mechanism is used to configure and adjust thresholds at runtime, which reduce the engineering effort. It can work with any number of sensors and users, and can be scaled up, too.

The concept has been tested using several third-party real-world datasets with different sensor deployments, user profiles, etc, and has been found to be effective.

Hopefully, we will see this and other research findings deployed in future smarthomes.

Smart, no doubt, but should be used smartly, too

The first reaction to these products and projects is amazement, followed by a sense of satisfaction, albeit peppered with a little fear. It is good to know that smarthome technologies enable people with disabilities to live on their own terms.

It is comforting that these technologies let mothers be at peace at their workplaces, while getting constant updates about the whereabouts and safety of their kids.

It is also nice that these technologies free up some of people's time and mindshare. Fortunately, these also save them a bundle by reducing energy consumption.

But, at what cost? When you find a free hour, you should listen to this National Public Radio debate on whether smart technology is making us dumb. It is available on www.npr.org/2015/05/20/407956931/debate-is-smart-technology-making-us-dumb

Hard to believe, but it could be true that too much of technology could reduce our ability to think and make decisions for ourselves. It is a boon for those who really need it, but others must exercise their discretion. You cannot have too much of a chocolate cake just because it is delectable. ●